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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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The overall goal of this project is to improve NASA risk estimates for space radiation-induced intestinal tumors. We are employing APC mouse models that have been successfully used to demonstrate radiation-induced intestinal cancer. The APC models have inactivation of one APC (adenomatous polyposis coli) gene and the mice develop intestinal tumors, adenomas. Radiation can inactivate the remaining APC gene and thus initiate increased intestinal tumor development. In addition to the APCmin model, we are currently employing the APC1638N model, which was generated by a targeted modification of the Apc gene at a position corresponding to aa 1638. These mice develop very few adenomas in small as well as in large intestine but post-radiation tumor numbers shows marked increases. The CDX2P Apc flox/+, is another APC mouse model we are using in our NSCOR studies, has one APC allele inactivated almost exclusively in the colon leading to adenomas and subsequent adenocarcinomas in the colon only. Data from these two mouse models will be combined with our previous studies on ApcMin/+ mice, a widely studied model for colorectal cancer to calculate risk estimates for NASA.

The strategy is to quantitatively analyze intestinal tumor incidence and grade following exposure to HZE ions, and/or high-energy protons of various energies, LET, and doses, and compare to gamma rays in the APC1638N and CDX2P Apc flox/+ models. Based on results in the Shay laboratory, efforts will also include protracted irradiation either with small graded doses given daily over 1 to 2 weeks or low dose rate, when available. We have initiated experiments to assess potential gender and age variation in tumor incidence with space radiation and also to characterize the effects of space radiation in triggering persistent stress responses and their potential effects on tumorigenesis. The plan is also to assess in these mouse models the relative effects of space radiation at other points in the carcinogenic process. In the future efforts will be put forward to delineate cancer progression-specific biological changes and genomic, proteomic and metabolomic signatures associated with radiation-induced tumorigenesis, as well as perturbations in tumor suppressor and oncogene-related pathways in radiation-induced tumors.

Since low-LET risk estimates are available for human populations, we will then be able to model the relative risk for space radiation. In particular, this ratio can then be extrapolated to human exposure to low-LET radiation to then make an estimate of risk for colorectal cancer in humans exposed to space radiation.

In order to answer the critical question of risk for colonic tumorigenesis after acute and chronic space radiation exposures, adequate in vitro human cell culture and animal models for CRC initiation and progression are needed. We are investigating the effects of high-LET particle and high-energy protons in colonic epithelial cells with respect to the nature of damage and consequent tumor induction using biological cancer endpoints and systems biology (genomics, proteomics and metabolomics) approaches. In addition, we propose to compare these results to the biological consequences of low dose rates and mixed fields of radiation which astronauts are likely to be continuously exposed. The study of low- and high-LET radiation on colonocytes gains more relevance with increasing age when premalignant adenomas and other colonic lesions such as flat lesions and microscopic aberrant crypt foci become more prevalent. Pre-existing undetectable genetic or chromosomal aberrations in astronauts in addition to de novo mutations resulting from long-term irradiation may increase the chances of colonic stem cell transformation.

There are the following three projects addressing the impact of the following: 1) Space radiation effects in mouse models of intestinal tumorigenesis ; 2) Space radiation effects in neoplastic progression events in normal human colonocytes ; 3) Space radiation effects in putative GI stem cells (beginning in year 3)

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Gastrointestinal tumors are frequent in the U.S., and American Cancer Society estimates that there will be 143,460 new cases of colorectal cancer (CRC) with 51,690 persons predicted to die of the disease in 2012. Colorectal cancer when considered in men and women is the second leading cancer killer in the United States (http://www.cdc.gov/). Considering the high frequency of colorectal tumors in the American population, an even small increase by space radiation could have a major impact on risk estimates and planning of future manned space missions. The effect of complex DNA lesions, which are produced by HZE radiation, on intestinal tumorigenesis may provide insight into mutagenic processes affecting genome integrity and colorectal carcinogenesis.

Considering the increased colorectal cancer incidence in A-bomb survivors and the novel characteristics of space radiation, cancer causation by space radiation could potentially be even greater. One general approach for risk assessment is to determine the relative biologic effectiveness (RBE) of various parameters for space radiation compared to terrestrial radiation exposures. For HZE ions there is more uncertainty with RBE for cellular parameters, such as chromosome aberrations and cell lethality, where values obtained from in vitro studies typically ranges from 2 to 11-fold. Previously we have reported the RBE of proton (1.06), 56Fe (1.25), 28Si (1.41), and 12C (0.99) for survival in female C57BL/6J mice. The rationale for this study is that acute toxicity for the planned radiation sources at the proposed energies were not available in mice.
Experiments with a solar particle event (SPE) simulation has been carried out using CDX2P Apc flox/+ mice as a model of colorectal adenoma-carcinoma progression to determine if the incidence or progression of colon cancer is increased after exposure to a total body dose of 2 Gy protons, a major component of space radiation. Mice (n=45 per group) were sacrificed as per approved protocol to determine if we observe an increase in the number of adenomas, adenoma-adenocarcinoma transitions, or the development of metastatic disease. Our data indicate that exposing these mice to a SPE simulation of 2 Gy of Protons over a 2 hours period significantly increases the incidence of polyps and carcinoma. SPE simulation also results in an overall decrease in survival compared to unirradiated mice. We will determine in the next review period if there are increases in polyp formation or carcinoma with both Fe and Si (single and fractionated doses).

Intestinal tumorigenesis, quantitatively and qualitatively, are studied in APC1638N/+ mice (6 to 8 weeks old). We are also determining age (6 to 8 wks vs. 20 to 24 wks old mice) and gender (male vs. female) effects of intestinal tumorigenesis. Recently, we performed intestinal tumor scoring for male APC1638N/+ that had received 0.1 and 1.4 Gy of 300 MeV/n 28Si radiation in a single exposure. Here we report that the frequency of intestinal tumors was significantly enhanced after 28Si exposure in both small and large intestine. We have also completed intestinal tumor scoring for female APC1638N/+ that had received 1.6 Gy of 1000 MeV/n 56Fe radiation in a single exposure. Significant increases in intestinal tumorigenesis were observed after 56Fe radiation exposure and tumor distribution in

Task Progress:	duodenum, jejunum, and ilium also showed increased tumors after 56Fe radiation. Importantly, we have utilized a genetic approach to determine if space radiation induced intestinal tumorigenesis could be inhibited. We generated APCMin/+;Wip1-/- mice and 6 to 8 wks old female mice were exposed to 1.6 Gy of 56Fe radiation. The Wip1 phosphatase, encoded by PPMID, is amplified in a variety of cancers and is p53-inducible but functions in a manner analogous to Mdm2 as a negative feedback factor with oncogenic properties. Here we report that 56Fe radiation induced intestinal tumorigenesis was significantly reduced in APCMin/+;Wip1-/- mice indicating that disruption of Wip1 could be an attractive target for cancer treatment and prevention. Task Progress on Human Colonic Epithelial Cells (HCECs): 1. We have developed an isogenic series of HCECs that are sensitized to cancer development. 2. We have conducted a soft agar-based short hairpin RNA (shRNA) screen within colorectal cancer (CRC) candidate driver genes (CAN-genes) using a karyotypically diploid hTERT- and CDK4-immortalized human colonic epithelial cell (HCEC) model and discovered that depletion of 65 of the 151 CAN-genes enhanced anchorage-independent growth in HCECs with ectopic expression of K-RasV12 and/or TP53 knockdown. 3. We now constructed an interaction map of the confirmed CAN-genes with CRC non-CAN-genes and screened for functional tumor suppressors. Remarkably, depletion of 15 out of 25 presumed passenger genes that interact with confirmed CAN-genes (60%) promoted soft agar growth in HCECs with TP53 knockdown compared to only 7 out of 55 (12.5%) of presumed passenger genes that do not interact. We have thus demonstrated a pool of driver mutations among the putative CRC passenger/incidental mutations, establishing the importance of employing biological filters, in addition to bioinformatics, to identify driver mutations. 4. We have demonstrated that Nrf2 activation by the synthetic triterpenoids, Bardoxolone methyl (BARD) and CDDO-ethyl amide, protect colonic epithelial cells against IR-induced damage in part by enhancing repair of DNA damage. BARD pre-treatment reduced the frequency of both G1 and S/G2 chromosome aberrations and increased the kinetics of appearance and disappearance of repairosomes (CtIP, Rad51, and 53BP1 foci) formation after IR exposure. BARD did not protect cells with Nrf2-knockdown. The 53BP1 promoter has three ARE sequences and Nrf2 directly binds to these AREs resulting in increased expression after BARD treatment. In addition, CDDO-EA provided before exposure to a lethal dose of total-body irradiation protected wild type mice from DNA damage and acute gastrointestinal (GI) toxicity and improved overall survival. These results demonstrate that Nrf2 activation by synthetic triterpenoids is a promising candidate target to protect the GI tract against acute IR exposure in vitro and in vivo.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
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